

CLOAK: A Privacy-Preserving Assistant for Online Social Media Users

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Abstract

Many people share personal details on platforms like Reddit under the assumption that a pseudonymous account keeps their identity safe. They do not realize that a profession mentioned in one subreddit, a hobby in another, and a neighborhood in a third can be pieced together to narrow down exactly who they are. Large language models have made this kind of cross-post profiling easy to do at scale. Yet existing tools offer little help: PII scanners catch names and emails but miss contextual cues, and even LLM-based privacy systems analyze only the current post in isolation, missing the cross-post combinations that actually compromise anonymity. We present CLOAK, a system that examines a user's full posting history alongside their draft comment and shows them the identification risk before they post. A pipeline of four LLM agents extracts a structured attribute profile from prior posts, identifies cross-post interactions that narrow the anonymity set, ranks risky phrases, and suggests minimal edits that preserve the user's intent. CLOAK is designed as an awareness tool for informed self-disclosure: rather than blocking or redacting content, it helps users understand the trade-offs and decide for themselves. In a technical evaluation on 368 Reddit profiles, CLOAK detected cross-post tipping points in 77% of cases while the status quo baseline detected none. In a user study ($N=20$), CLOAK was unanimously selected as the most helpful tool (trust: 6.5/7 vs. 1.5/7 for PII scanning), and every participant reported changed posting behavior.

CCS Concepts

• **Security and privacy** → **Privacy protections; Usability in security and privacy**; • **Human-centered computing** → *Interactive systems and tools*.

Keywords

privacy, cross-post profiling, quasi-identifiers, LLM agents, Reddit, informed self-disclosure

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1 Introduction

Pseudonymous platforms like Reddit give millions of people a space to discuss things they would never attach to their real name [4, 26]: mental health struggles, workplace frustrations, relationship problems, financial worries. Users rely on the separation between their username and their real identity to speak freely [14], and communities rely on that honesty to function. But this separation is more fragile than most people realize.

Consider a nurse who vents about burnout in *r/nursing*. A few weeks later she mentions her salary in *r/personalfinance*. She talks about her golden retriever in *r/dogs*, her vegetable garden in *r/gardening*, and a drug shortage at her hospital in *r/medicine*. None of these posts contain her name, email, or phone number. A PII scanner would call each one safe. But taken together, they describe an emergency nurse with eight years of experience, a specific salary range, a charge nurse title, a pet breed, and a metro area. When she then drafts a comment in *r/Divorce* mentioning the nursing school she attended and her six-year-old child, the combination narrows the field to one person. A colleague, a supervisor, or an ex-partner reading both the nursing posts and the divorce comment could connect the dots without any technical skill.

Until recently, this kind of cross-post profiling required expensive manual work [31] that few adversaries would bother with. Large language models have changed that. Staab et al [15, 39, 44] showed that GPT-4 can infer personal attributes like location, age, and income from Reddit comment histories with up to 85% accuracy, at a fraction of the cost of human analysts. Du et al [12] built AutoProfiler, a system that pulls together a user's public posts into a detailed personal profile, recovering attributes like health conditions and relationship status even from text that had been run through commercial anonymization tools. Lermen et al [22] showed that LLMs can go further, linking pseudonymous accounts to real identities across platforms. Anyone with access to an LLM can now



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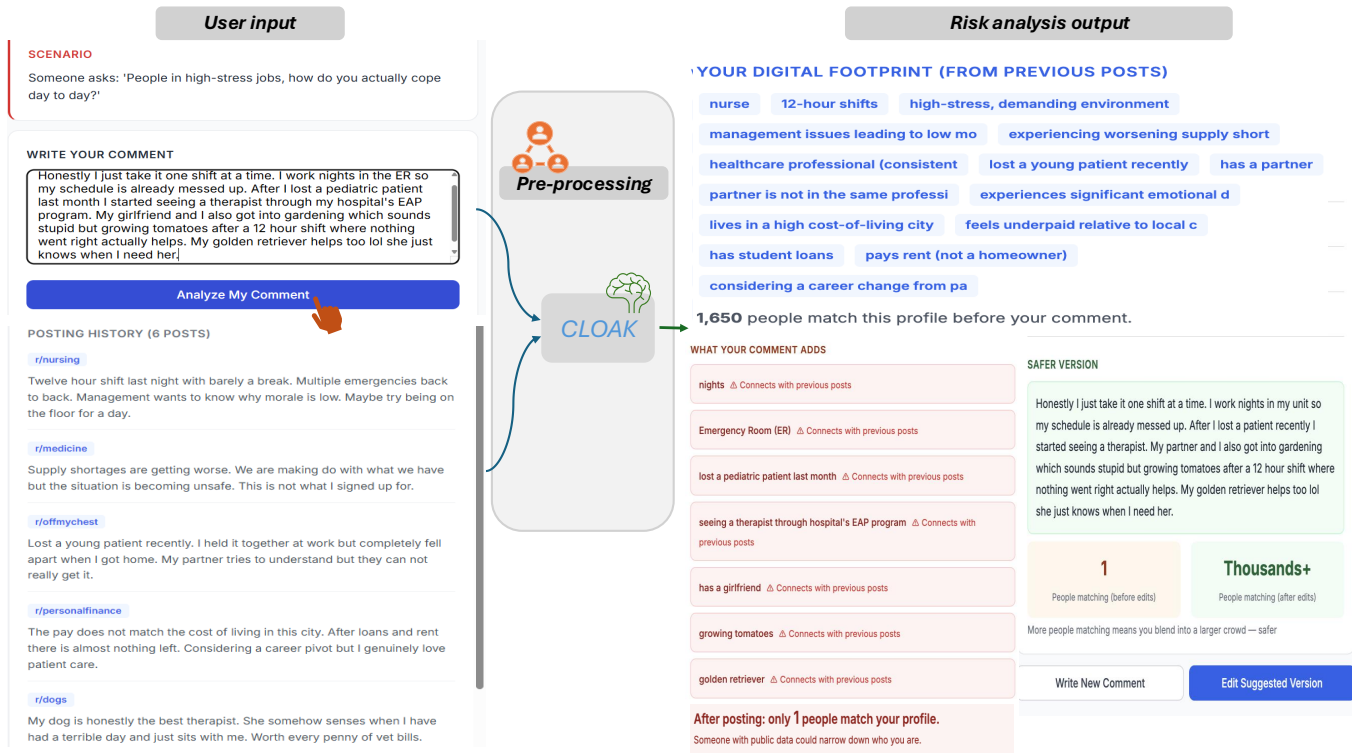


Figure 1: CLOAK analyzes how a new comment interacts with a user’s posting history to create identification risk. (1) A healthcare worker writes a comment about coping with stress, with their posting history across Six subreddits shown on the left. (2) CLOAK builds a digital footprint from previous posts (1,650 matching people) and identifies seven phrases in the draft that connect with the history. (3) After posting, only one person matches. CLOAK explains which phrases carry the most risk and suggests a lower-risk version that expands the matching population back to thousands.

do what used to take a team of investigators, but none of this work offers users a way to protect themselves.

The privacy tools that do exist are not designed for this threat. PII scanners like Presidio [25, 29] look for names and emails. LLM-based anonymizers [40] can rewrite a single comment to reduce what an attacker can learn from it. But both approaches analyze one text at a time and never look at what the user has posted before. A comment that seems safe on its own can be the one detail that, combined with months of prior posts, gives the user away [27]. In our user study, participants rated their general privacy concern at the maximum (7/7) but their awareness of cross-post identification at the midpoint (4/7) [17, 49]. They care about privacy, but they do not know how their anonymity actually breaks down, and the tools they have access to tell them their posts are safe when they are not.

We built CLOAK to fill this gap. CLOAK takes a user’s posting history and a draft comment and shows them how the two interact to create identification risk (Figure 1). It points out which phrases in the draft are the problem, explains why they are risky in the context of previous posts, and suggests small changes that keep the meaning of the post intact while making the user harder to identify. CLOAK does not block or redact anything. Following the principle of informed self-disclosure [11, 20], it gives users the information they need to make their own decision about whether to post.

We designed and evaluated CLOAK in three stages. First, we built a pipeline of four LLM agents (Gemini 2.5 Pro and Flash [43]) that breaks the analysis into six steps: building a profile from the user’s history, extracting new attributes from the draft, computing how much the draft narrows the user’s anonymity, ranking which phrases contribute the most risk, suggesting edits, and verifying that the edits work. Along the way, we found that beyond the standard PII and SPI categories used in prior work [12], there is a large class of attributes we call quasi-identifiers. These are details like career transitions, niche hobbies, and unique life events that are harmless alone but become identifying when combined across posts. In our dataset, they make up the majority of identity-relevant information, yet no existing tool flags them.

Second, we tested CLOAK on a dataset of 368 Reddit profiles that includes constructed personas, real Reddit users from r/IAmA, and an extension of the SynthPAI benchmark [51]. We compared it against six baselines, from Presidio to carefully written single-prompt approaches. CLOAK detected cross-post tipping points in 77% of cases (89% on curated profiles). The strongest single-post baseline detected none.

Third, we ran a within-subjects user study where 20 participants tried three tools side by side on the same Reddit comment: a PII scanner, an AI privacy scanner that looks at only the current post,

and CLOAK. All 20 selected CLOAK as the most helpful, with trust ratings of 6.5/7 compared to 1.5/7 for PII scanning. Every participant said the study changed how they think about posting.

In sum, the paper makes four contributions: (i) The identification of **quasi-identifiers** as a large and previously unaddressed part of cross-post disclosure risk, along with an annotated evaluation dataset of 368 Reddit profiles; (ii) The design and implementation of **CLOAK**, a pipeline of four LLM agents that detects cross-post identity exposure, ranks risky phrases, and suggests edits that reduce risk while keeping the post’s meaning; (iii) A **technical evaluation** showing that CLOAK detects cross-post tipping points in 77% of evaluated profiles and that 62.8% of profiles end up with fewer than 100 matching candidates after a single additional comment; and (iv) A **within-subjects user study** ($N=20$) in which CLOAK was unanimously chosen as the most helpful tool, with a high System Usability Scale score of 81.6.

2 Related Work

2.1 User Profiling and De-anonymization

People have always assumed that a username is enough to keep their identity hidden on platforms like Reddit [34]. That assumption was never as safe as it felt. Sweeney showed over two decades ago that just three demographic attributes (ZIP code, gender, date of birth) could uniquely identify 87% of Americans in public databases [41]. Still, the practical risk stayed low for most people, because piecing together scattered online posts required manual work that few adversaries would bother with.

Large language models have removed that barrier. Staab et al [39] gave Reddit comment histories to GPT-4 and asked it to infer location, age, income, and sex. The model got the right answer 85% of the time, at 100× less cost than human annotators. Standard anonymization could strip out names and addresses, but the model picked up on cues no anonymizer would catch: regional slang, cultural references, how someone talks about time. Yukhymenko et al [51] built on this with SynthPAI, a dataset of over 7,800 Reddit-style comments that confirmed the same profiling results across 18 LLMs. Du et al [12] took the next step with AutoProfiler, a system that assembles a user’s public posts into a structured personal profile, recovering attributes like health conditions and relationship status even from text that had been run through commercial anonymization tools. Zheng et al [53] took a complementary approach, developing probabilistic reasoning methods that estimate k -anonymity-based privacy risks directly from user posts, providing a formal framework for quantifying how identifiable a user is from their public content. Lermen et al [22, 37] showed that LLMs can go further still, linking pseudonymous accounts to real identities across platforms. Li [23] demonstrated that agentic LLMs can autonomously re-identify participants from interview transcripts by chaining multiple reasoning steps. Ko et al [16] further showed that even weak cues such as writing style and topical preferences are sufficient for inference-driven de-anonymization when processed by modern language models.

All of this work sits on the attack side of the problem. Each paper shows what an adversary can learn from scattered posts. None of them gives users a way to view/address this risk before they post.

2.2 Privacy-Preserving Text Anonymization

Given how well LLMs can profile users from their posts, the obvious next question is whether text can be rewritten to stop them. Anonymization has traditionally meant removing explicit identifiers like names, phone numbers, and email addresses, but the newer attacks work on other aspects [25, 28]. Tools like Microsoft Presidio [29] use named entity recognition and pattern matching to find and mask direct PII. They are fast and widely deployed, but they only catch what is stated outright. Staab et al [39] showed this gap directly: after running Reddit comments through Azure’s anonymizer, LLMs could still infer location and age from cues the tool never touched.

Newer approaches use LLMs themselves as anonymizers. Staab et al [40] set up a loop where one LLM rewrites text while another tries to infer personal attributes from the rewritten version, repeating until the attacker fails. This outperformed commercial anonymizers on both privacy and utility. Other work has explored differential privacy for text [18] and k -anonymity style redaction [10], but these often make the text harder to read. Each of these methods analyzes a single text in isolation. What came before, i.e., months of posts across different subreddit comments, is invisible to them.

2.3 Privacy Tools and Informed Disclosure

Even when effective privacy tools exist, they only help if users understand what they are being warned about and know what to do next [36]. Research on privacy nudges [2] has shown that small, well-timed interventions can reduce over-sharing, but the design matters: vague warnings are easy to dismiss, while specific feedback tied to a concrete risk changes user behavior [48].

Research on self-disclosure explains why this problem runs so deep. People consistently underestimate how much they reveal over time [4, 13, 52]. Each post feels harmless on its own, and users rarely think about how their posts interact with each other across weeks or months [1, 32]. Kang et al [14] describe this as “benign disinhibition”: the feeling of safety under a pseudonym leads people to share more, not less.

A few recent systems tackle this directly. Monteiro et al [30] built Imago Obscura, an AI co-pilot for image privacy that flags risks in photos and recommends ways to address them. In a study with 15 participants, users strongly preferred being informed over being constrained: they wanted to see the risk and decide for themselves.

Most directly relevant are the works of Krsek et al [19, 20], who studied how to present re-identification risk to Reddit users. They tested five designs for population risk estimates with 44 participants and found that showing the risk raised awareness but also caused anxiety when users could see the risk but did not know how to reduce it. Their four recommendations, i.e., explain how disclosures could be exploited, show how the estimate was calculated, offer concrete suggestions, and use plain language, directly shaped how we designed CLOAK. Their work however only assesses one disclosure at a time and does not model how attributes accumulate across posts. Recognizing this limitation, they call for future work that reasons over posting history, cross-platform footprints, and other publicly accessible information. CLOAK directly addresses this open question. CLOAK is the first system that examines a user’s

full posting history and shows them, before they post, how a new comment interacts with everything they have already shared.

3 CLOAK

CLOAK takes two inputs: a user’s existing Reddit posts (already public) and a draft comment (about to be published). It analyzes how the draft interacts with the posting history to create identification risk, and produces a report showing what an adversary could learn from the combination, which phrases in the draft are responsible, and how to change them without losing the utility of the post.

3.1 Threat Model

CLOAK is designed to protect against a generic adversary: anyone who can read the user’s public posting history and search public sources such as professional directories, LinkedIn, or institutional websites. The adversary does not need prior knowledge of the user. What CLOAK estimates is how rare the user’s combination of attributes is in the general population. The candidate set represents how many people would match the same combination if an adversary searched the public databases. A candidate set of one means the combination is specific enough that a search can match exactly one person. An example walkthrough (Appendix C) demonstrates this end to end. Interpersonal adversaries such as colleagues or ex-partners pose a greater risk because they already have context a stranger would need to search for. CLOAK’s estimates reflect the harder case of a stranger with no prior knowledge, so a small candidate set means the user is identifiable to strangers and certainly to anyone who already knows them. We do not model nation-state adversaries or platform insiders with access to private data.

3.2 Dataset

No existing dataset captures how identity risk builds up across posts on pseudonymous platforms. Benchmarks like SynthPAI [51] provide attribute annotations for individual comments, but they do not model the cross-post interactions that make someone identifiable. We therefore created a dataset where each entry has a posting history, a draft comment, and ground truth about whether the draft creates a tipping point. We built this dataset in three tiers.

Tier 1: Real Reddit profiles (20 profiles). We collected posting histories from 20 real Reddit profiles sourced from *r/IAmA*, where people voluntarily verify their identity through photo verification or linked professional profiles. For each profile, we collected posts from the user’s non-AMA subreddits and removed identifiable information to serve as the posting history input, while the verified identity from their AMA served as the ground truth. These profiles exhibit the messiness of real posting behavior: ambiguous language, sarcasm, implicit disclosure, and attributes scattered across months of activity. This makes Tier 1 the most challenging evaluation tier, and it also served as the foundation for the next tier.

Tier 2: Constructed profiles (50 profiles). By studying the posting patterns in the 20 real profiles, we identified five common archetypes: graduate student, tech professional, educator, health-care worker, and young professional. We then created 50 synthetic Reddit profiles modeled after these patterns, each with 4 to 8 posts spread across 3 to 6 subreddits and one draft comment that introduces new identifying details. We chose these archetypes because

they represent different kinds of privacy risk: a graduate student in a small research group is identifiable in very different ways than a nurse working night shifts at a large hospital. For each profile, we manually annotated every attribute in the history, every new attribute in the draft, the specific cross-post interactions between them, and whether the draft creates a tipping point. These annotations serve as ground truth for the baseline comparison.

Tier 3: SynthPAI extension (298 profiles). To test at scale, we extended SynthPAI [51] using a leave-one-out method. For each of the 298 profiles, we identified the comment with the most human-confirmed attribute disclosures and designated it as the critical draft. All remaining comments form the posting history. This annotation comes entirely from SynthPAI’s existing human review labels, so it introduces no circularity from our own pipeline. We split these profiles by the strength of the critical draft: 142 have two or more confirmed attributes (strong), 139 have exactly one (weak), and 17 have no identifiable post at all, included to test whether CLOAK avoids flagging profiles with no cross-post risk. Full details, including example profiles from each tier, appear in Appendix A.

3.3 Attribute Taxonomy

To reason about what makes someone identifiable, CLOAK needs a way to categorize the kinds of information people share online. Prior work on user profiling [12] classifies extracted attributes into PII (directly identifying information like name, age, location) and SPI (sensitive information like health conditions or political views that causes harm if linked to a real identity). We found that these two categories miss the largest source of cross-post risk, so we added a third category.

QI (Quasi-Identifiers) includes attributes that are neither identifiable nor sensitive alone, but become identifying when combined across posts. Career transitions, niche hobbies, unique life events, specialized expertise. A post mentioning “I used to work as a paramedic” is harmless. Combined with “I write fiction professionally” from a different subreddit and “I grew up in a small town in the Middle East” from a third, the intersection can dramatically narrow the candidate set.

This third category turns out to be the largest. Across the 70 manually annotated profiles in Tiers 1 and 2, QI makes up 55.5% of all identity-relevant attributes, compared to 23.0% for PII and 21.5% for SPI (Table 5 in Appendix A). More than half of what makes someone identifiable is information that no existing tool flags.

3.4 Pipeline

CLOAK breaks cross-post analysis into six steps, each handled by one of four LLM agents (Figure 2; agent configuration in Appendix E, Table 13). None of the agents are fine-tuned; they operate through few-shot prompting [8] with carefully written system and task prompts. The full prompts are provided in Appendix E. The following two design decisions shaped this pipeline.

Separating history from draft. Rather than feeding all posts into a single call, CLOAK first builds an attribute profile from the user’s existing history, and then evaluates the draft against that profile. This reflects how the problem actually works: the history is already public and cannot be taken back, but the draft can still

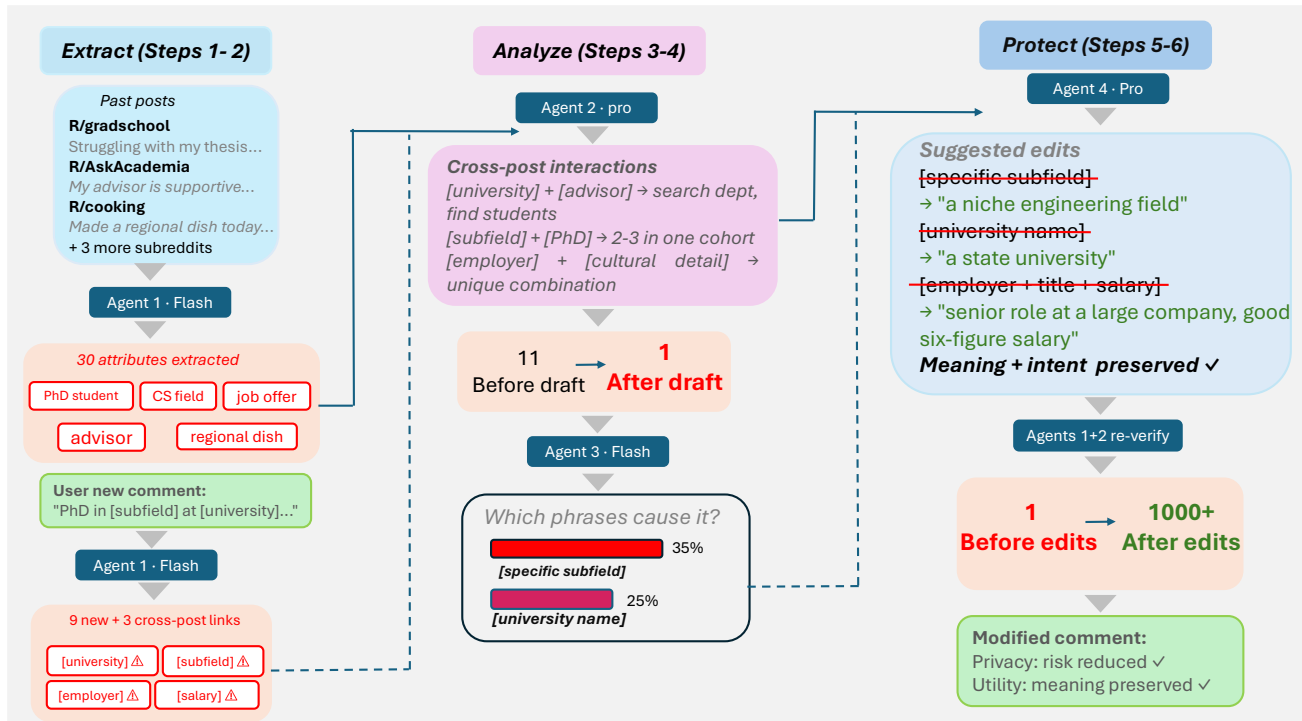


Figure 2: CLOAK’s six-step pipeline, illustrated with an anonymized graduate student profile. Extract (Steps 1-2): Agent 1 builds a 30-attribute profile from the posting history, then scans the new user comment and finds 9 new attributes with 3 cross-post links. Analyze (Steps 3-4): Agent 2 identifies how new attributes interact with the history and estimates the candidate set narrowing (11 people → 1). Agent 3 ranks which phrases contribute most to the risk. Protect (Steps 5-6): Agent 4 suggests minimal edits that reduce the cross-post risk while preserving meaning and voice. Agents 1 and 2 re-check the edited draft, estimating the candidate set expands back to over 1,000.

be changed. It also means the history profile is computed once and reused whenever the user checks a new draft.

Using multiple agents instead of one. In early experiments, we tried giving a single LLM the entire task: extract attributes, estimate population narrowing, rank phrases, and suggest edits, all in one prompt. The model would default to arbitrary, unverified estimates rather than reasoning through each narrowing step with supporting evidence, and the suggested edits stripped away the user’s intent rather than preserving it. When we split the work into four agents, each with its own role and examples, both problems went away. Separate agents also let us pick the right model for each step: the reasoning-heavy steps (risk analysis, edit suggestion) run on Gemini 2.5 Pro, while the faster classification steps (attribute extraction, phrase ranking) run on Gemini 2.5 Flash at roughly one-fifth the cost.

We walk through each step using one of our evaluation personas, a graduate student who has posted across six subreddits about their research, career plans, cooking, gaming, and local community.

Step 1: Profile Build (Agent 1, Gemini Flash). Takes the user’s history posts as raw text and processes them in a single call. The agent extracts every identity-relevant attribute, returning a structured profile where each attribute is tagged with its value, source

post, subreddit, category (PII, SPI, or QI), confidence score, and risk level. For the graduate student, this step found 30 attributes from six posts, including “PhD student” from a research subreddit, “has an advisor from a similar cultural background” from an academic advice post, “received a startup job offer” from a career subreddit, and “cooks a specific regional dish” from a cooking post.

Step 2: Draft Extraction (Agent 1, Gemini Flash). Takes the draft comment along with a summary of the existing profile. The agent reads the draft in the context of what is already known, picks out attributes that are new, and flags cross-post interactions: cases where something in the draft connects with something from the history to create risk. The graduate student’s draft mentioned a specific university, which alone is unremarkable, but combined with the advisor reference from the history, it lets someone search that university’s faculty directory and find the advisor’s students.

Step 3: Risk Analysis (Agent 2, Gemini Pro). Estimates how many people in the world match the user’s attribute profile, in two conditions: before the draft (history only) and after the draft (history plus new attributes). The agent builds a narrowing chain using public data sources like the U.S. Census and Bureau of Labor Statistics, and identifies the tipping point: the draft attribute that causes the biggest drop. For the graduate student, the candidate set

dropped from 11 people to 1. This does not mean the system found a real person. It means the combination of attributes is specific enough that, statistically, only about one person in the population would match. If someone with that exact profile existed, they could be found through a public directory or LinkedIn search.

Step 4: Phrase Ranking (Agent 3, Gemini Flash). Assigns a risk percentage to every identifying phrase in the draft based on how much it contributes to the narrowing through its interaction with the history. “PhD in [specific subfield]” was ranked at 35% not because it is dangerous on its own, but because it combined with the existing “PhD student” attribute to narrow the pool to two or three people in one department.

Step 5: Edit Suggestion (Agent 4, Gemini Pro). Suggests the smallest changes that would break the cross-post link while keeping the user’s voice and meaning. Edits follow a preference order: generalize first (a specific university becomes “a state university”), fuzz numbers second (a specific salary becomes “a good six-figure salary”), rephrase third, and remove only as a last resort. Each suggestion explains what the user gives up and what they keep.

Step 6: Verification (Agents 1 + 2). Re-runs Steps 2 and 3 on the edited version to check whether the edits reduced the cross-post risk. For the graduate student, the edited draft expanded the matching population from 1 person to thousands.

The full pipeline makes seven LLM calls per profile. The first run takes about 285 seconds. After that, the history profile is cached, and checking a new draft takes 120 to 150 seconds. The estimated cost is about \$0.04 per analysis at current Gemini API pricing. A detailed cost breakdown appears in Appendix E.

3.5 User Interface

We designed CLOAK’s interface around a simple idea: the best way to show someone why cross-post analysis matters is to let them see what existing tools miss. The interface presents three tabs, each representing a different level of privacy analysis (Appendix D)

The **PII Scanner** tab runs Microsoft Presidio on the draft. It checks for direct identifiers: names, email addresses, phone numbers. For the graduate student’s comment, this tab reports “No personally identifying information detected. Your comment appears safe to post.” A note below explains that the scanner only looks for direct identifiers and cannot see how the comment connects with previous posts.

The **AI Privacy Scanner** tab runs a three-round loop inspired by recent anonymization work [40]: one model infers attributes from the draft, another rewrites it to remove them, and a third checks that the rewrite still makes sense. This produces a lower-risk version of the comment, but since it never sees the posting history, it misses all cross-post interactions. A notice at the bottom says so: “This analysis only examined your current comment. It cannot detect how this comment combines with your previous posts to narrow down your identity.”

The **CLOAK** tab shows the full analysis in five parts:

(a) *Your Digital Footprint*. Every attribute accumulated from the posting history, shown as labeled tags. This makes the invisible profile visible for the first time.

(b) *What Your Comment Adds*. New attributes from the draft, each marked with a warning if it connects to something from history.

(c) *How This Could Identify You*. The specific cross-post interactions, written in plain language. For example: “Your mention of [university] combined with your earlier post about your advisor lets someone search that university’s faculty page and find the advisor’s students.”

(d) *How To Protect Yourself*. Risk percentages for each phrase, followed by a lower-risk version with all suggested edits applied.

(e) *Before/After*. The candidate set before edits and after, with explanation: “More people matching your profile means you are harder to single out. This reduces your risk but does not eliminate it.”

CLOAK never blocks a post. Users can post as written, apply the suggested edits, or use the risk information as they see fit.

3.6 Personas

Rather than asking participants to connect their own Reddit accounts, which would raise ethical concerns and limit experimental control, we designed five personas for the user study. Each persona has a curated posting history across multiple subreddits and a scenario prompt that naturally invites a detailed comment.

The **Graduate Student** has posted across research, career, academic advice, cooking, gaming, and local community subreddits, revealing a PhD student with an identifiable advisor and cultural background markers. The **Tech Professional** has posted across career, finance, parenting, and hobby subreddits, revealing a recently promoted employee with limited internal mobility and layoff anxiety. The **Healthcare Worker** has posted about shift schedules, patient care, and professional certification. The **Educator** has posted about school system challenges and classroom dynamics. The **Young Professional** has posted about early career decisions, housing, and work-life balance. Each persona covers a different kind of privacy risk, and each participant was assigned one persona counterbalanced with condition order. Full persona descriptions, including posting histories and scenario prompts, are in Appendix D.

4 Evaluation

We evaluate CLOAK along 2 axes: does the pipeline produce accurate cross-post risk analysis, and do users find it valuable when deciding what to post? We conduct a technical evaluation on the 368-profile dataset described in Section 3.2, and also a within-subjects user study with 20 participants. Figure 3 summarizes the technical findings and user impact; Figure 4 presents the user study results.

4.1 Technical Evaluation

4.1.1 Baselines. Cross-post identity exposure detection is a new task, so no existing system was designed for it. We adapted six methods from related areas to serve as comparisons. Three receive only the draft comment: a *PII Scanner* (Presidio [29]), a *Single-Post LLM* (Gemini Pro analyzing the draft with CLOAK’s full prompt), and a *Zero-Shot LLM* (a minimal “is this risky?” prompt). Three receive the full posting history as well: a *Batch LLM* (all posts concatenated without structure), a *Monolithic LLM* (one prompt covering all of CLOAK’s tasks at once), and an *Expert Prompt* (a carefully written single prompt with definitions, examples, and step-by-step instructions). Full baseline details appear in Appendix B.

4.1.2 Detection results. CLOAK detected cross-post tipping points in 77% of cases overall and 89% on the 70 curated profiles (Tiers 1 and

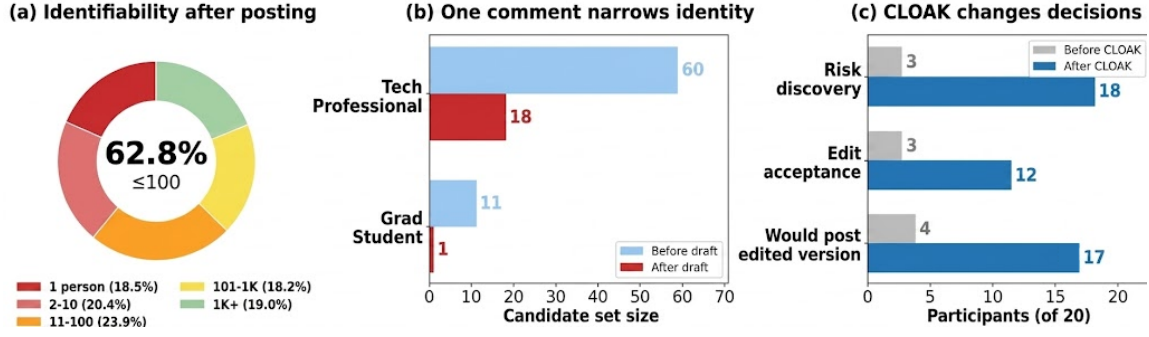


Figure 3: Evidence and impact. (a) After posting, 62.8% of profiles are narrowed to fewer than 100 candidates; 18.5% to one person. (b) Two concrete examples: one comment drops the graduate student from 11 candidates to 1, and the tech professional from 54 to 18. (c) After seeing CLOAK’s analysis, participants discovered new risks (3→18), accepted edits (3→12), and were willing to post the lower-risk version (4→17).

Table 1: Baseline comparison on the 70 curated profiles. *Det.* = percentage of tipping points correctly detected. *Int. recall* = fraction of ground-truth cross-post interactions found. [†]Flags every post as risky but identifies zero cross-post interactions.

Method	Det.	Int. recall	Scope	Edits
PII Scanner	0%	0.00	Single post	No
Zero-Shot LLM	100% [†]	0.00	Single post	No
Single-Post LLM	0%	0.00	Single post	No
Batch LLM	14%	0.00	All posts	No
Monolithic LLM	29%	0.80	All posts	Yes
Expert Prompt	50%	0.90	All posts	Yes
CLOAK	89%	0.85	History + draft	Yes

2). On the 20 real r/IAmA profiles, CLOAK correctly detected 16 of 20 tipping points. On the 50 constructed profiles, it detected 46 of 50. The lower rate on real profiles reflects the messiness of real posting behavior: ambiguous language, sarcasm, and implicit disclosures make tipping points harder to detect. Importantly, CLOAK produced few false alarms on both tiers, rarely flagging a safe draft as risky. We frame a larger evaluation on real accounts as future work. The profiler extracted about 30 attributes per profile from the posting history and identified an average of 3 new attributes per draft. To put the candidate set numbers in perspective: before a user writes their draft, their accumulated posting history already narrows the pool of possible matches from the general population to roughly 1.3 million people on average. After adding just one comment, that number drops by 93.6% to about 81,000. For 62.8% of profiles, the candidate set fell below 100 people, and for 18.5%, it reached exactly one person (Figure 3a).

To be clear, CLOAK does not search for or identify a real individual. It estimates how many people in the population would share the same combination of attributes, using public data sources like the U.S. Census [46] and Bureau of Labor Statistics [45].

Table 1 compares CLOAK to the six baselines on the curated profiles. The single-post methods are the most revealing. The Zero-Shot LLM flags every comment as risky, but since it never sees the

history, it cannot explain *why* or point to any cross-post interaction. The Single-Post LLM, which represents the strongest tool a user could use today to check a single comment, detected 0% of tipping points. It looked at the comment, found nothing wrong, and said it was safe to post.

The baselines that receive the full history do better at finding individual interactions. The Expert Prompt found 90% of them, slightly more than CLOAK’s 85%. But turning those individual findings into a correct overall risk judgment is where single-prompt approaches break down. The Expert Prompt detected only 50% of tipping points, and the Monolithic prompt only 29%. When one LLM tries to extract attributes, estimate populations, rank phrases, and suggest edits all at once, the different tasks interfere with each other. CLOAK’s four separate agents, each focused on one step, achieve 89% detection while maintaining strong interaction recall.

4.1.3 Output quality. An independent LLM judge [54] (Gpt 5 [38]) rated CLOAK’s output across seven quality dimensions on a 1-5 scale, evaluating 284 profiles which were labeled as having tipping points by our system. The overall average was 4.50 out of 5. The parts of the output that users interact with directly scored highest: phrase accuracy (are the right words flagged?) at 4.91, edit quality (is meaning preserved in the rewrites?) at 4.86, and actionability (can a user act on the suggestions?) at 4.84. Profile quality scored lowest at 3.53, reflecting the difficulty of parsing informal Reddit text where sarcasm, hypotheticals, and inside jokes can look like genuine disclosures to an LLM. Full quality scores appear in Appendix C.

4.1.4 Do the edits help? We measured whether CLOAK’s suggested edits actually reduce what an attacker can learn. We ran a simulated re-identification attack on each profile twice, once with the original draft and once with CLOAK’s edited version, and compared the number of personal attributes the attacker could infer. The results split into three groups. For 32.8% of profiles, the draft was the tipping point: it was the comment that made the user identifiable, and CLOAK’s edits reduced the number of inferable attributes by an average of 13.4%. The effect was even stronger for the most sensitive attributes: among high-confidence inferences, those an attacker would be most certain about, the reduction was 18.2%. For 35.9% of profiles, the posting history alone was already enough to identify

the person, so editing the draft comment could not change the outcome. For the remaining 31.3%, edits slightly increased what the automated attacker could pick up, because rephrasing sometimes makes implicit information more explicit to a machine even when the text reads safer to a human. Full privacy-utility analysis appears in Appendix C.

4.1.5 Are the risk estimates grounded? CLOAK’s risk analysis produces two outputs that users see directly: the attributes extracted from their posts and the candidate set estimate. If the extracted attributes are wrong, or if the narrowing chain does not follow from them, the risk estimate becomes unreliable. Ground-truth candidate lists are impractical to construct at scale because verifying the true population matching a specific attribute combination would require access to databases that do not exist in combined form. We therefore validated CLOAK’s estimates using a human-as-judge protocol [47], consistent with prior work in this area that relies on human verification of model inferences [39, 51], in which two annotators independently evaluated the 70 curated profiles on two dimensions using a 5-point Likert scale. Because the ratings are ordinal, we report quadratic-weighted κ [9], which gives partial credit for adjacent scores rather than treating every disagreement as equally severe.

Factuality assessed whether the attributes CLOAK extracted were actually present in the posts, and whether the population figures cited in the narrowing chain were reasonable rather than fabricated (e.g., scoring lower if CLOAK listed “registered nurse” when the post only said “I work in healthcare”). The mean factuality scores were 4.29 (Annotator 1) and 3.69 (Annotator 2), with an overall mean of 3.99 and substantial agreement [21] (quadratic-weighted Cohen’s $\kappa = 0.76$). **Relevance** assessed whether the narrowing steps followed logically from the attributes and soundly led to the reported candidate set (e.g., scoring lower if CLOAK jumped from “nurse in the US” to “one person” without intermediate steps). The mean relevance scores were 4.09 (Annotator 1) and 4.33 (Annotator 2), with an overall mean of 4.21 and substantial agreement (quadratic-weighted Cohen’s $\kappa = 0.81$). Annotators could verify candidate counts by searching public directories and databases online.

Relevance scored higher because narrowing logic is easier to verify than attribute extraction. Factuality scored lower because informal Reddit text contains sarcasm, hypotheticals, and exaggeration that can look like genuine disclosures to an LLM, consistent with the profile quality score of 3.53 in Table 8.

4.2 User Study

4.2.1 Design. We ran a within-subjects lab study comparing three tools, each representing a different level of privacy analysis:

Condition A: PII Scanner. Presidio-based detection on draft comment only. This is the most common type of privacy tool available to users today. It checks for names, emails, and phone numbers.

Condition B: AI Privacy Scanner. An LLM-based tool that analyzes the current comment in isolation using a three-round loop. It finds inferable attributes and suggests edits, but never looks at the posting history.

Condition C: CLOAK. The full cross-post analysis described in Section 3, which reads the entire posting history alongside the draft.

Condition order was counterbalanced across six Latin-square orderings. Each participant tried all three tools on the same comment, so they could compare the results directly.

4.2.2 Participants. We recruited 20 participants (15 male, 5 female, aged 18–34) from a university population. Participants came from multiple fields including computer science, mechanical and aerospace engineering, and business, and included students, university staff, and an industry professional. Fourteen were occasional Reddit users, two posted weekly, three had never used Reddit, and one did not report usage. Eight had used throwaway accounts and nine had deleted posts for privacy reasons before.

Before seeing any tool, we measured baseline privacy awareness with three 7-point Likert items ($N = 19$; one participant’s form was incomplete; full questions in Appendix D). General privacy concern was high (D1 Mdn = 7): most participants care a lot about privacy. Awareness of cross-post identification was moderate (D2 Mdn = 4): they had some sense that posts could be connected, but had not thought carefully about how comments across different subreddits could be combined to identify them.

4.2.3 Procedure. Sessions lasted 30 to 45 minutes. After informed consent and a demographics questionnaire, each participant was assigned one of five personas and shown its posting history. For each tool, they read a scenario prompt, wrote or reviewed a comment, and then saw the tool’s output. After each tool, they rated perceived risk (how risky does this comment feel?), confidence (how sure are you?), understanding (do you understand the risk?), and willingness to post (would you post this as-is?). After experiencing all three tools, they completed a tool comparison questionnaire, feature ratings for CLOAK, the System Usability Scale [3], and an exit interview lasting 5 to 10 minutes. Each persona was assigned to four participants, counterbalanced with condition order. Full study materials appear in Appendix D.

4.3 User Study Results

4.3.1 Tool preference and trust. All 20 participants selected CLOAK as the most helpful tool. This was unanimous across all five personas and all six condition orderings: no matter which persona a participant used or which order they saw the tools in, CLOAK was the top choice.

Trust ratings told the same story (Table 2, Figure 4a). On a 7-point scale, the PII Scanner received a median trust of 1.5, the AI Privacy Scanner 3, and CLOAK 6.5. A Friedman test confirmed significance ($\chi^2(2) = 34.1, p < .001$). Pairwise Wilcoxon tests with Bonferroni correction showed that CLOAK received significantly higher trust than both the PII Scanner ($Z = -3.83, p < .001, r = .88$) and the AI Privacy Scanner ($Z = -3.84, p < .001, r = .88$). Even the participant who trusted CLOAK the least gave it a 5 out of 7. Neither baseline reached a median above 3.

4.3.2 Analysis quality. Participants rated CLOAK and the AI Scanner on the same 7-point scale (Table 12 in Appendix D, Figure 4b). The PII Scanner does not produce risk analysis or suggested edits, so these dimensions do not apply to it. The AI Scanner, which analyzes only the current comment, scored a median of 4 for usefulness, 2.5 for edit quality, and 2.5 for clarity. CLOAK, which analyzes the draft against the full posting history, scored 7 on all three. All three

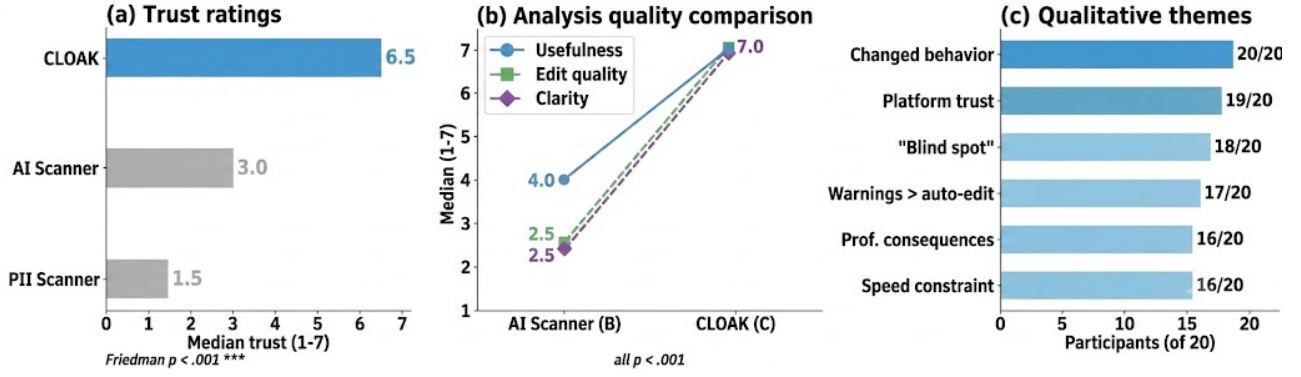


Figure 4: User study results ($N=20$). (a) Trust ratings: CLOAK (Mdn = 6.5) outperforms both baselines (Friedman $\chi^2(2) = 34.1$, $p < .001$). (b) Analysis quality: AI Scanner (B) vs. CLOAK (C). Usefulness, edit quality, and clarity all significantly higher for CLOAK (all $p < .001$). (c) Six qualitative themes from interviews; all endorsed by at least 16 of 20 participants.

Table 2: Trust ratings (1 = No trust, 7 = Full trust). $N=20$ for PII and CLOAK; $N=19$ for AI (one form invalid).

Tool	Mdn	Mean	SD	Range
PII Scanner	1.5	1.95	1.19	1–5
AI Privacy Scanner	3	2.89	1.56	1–6
CLOAK	6.5	6.40	0.68	5–7

Table 3: CLOAK feature usefulness (7-point Likert, $N=20$).

Feature	Mdn	Mean	SD	≥ 6
Cross-post interaction (Q5a)	7	6.75	0.44	20/20
Digital footprint (Q5e)	7	6.55	0.69	18/20
Phrase-level risk (Q5b)	7	6.50	0.69	18/20
Consequences (Q5d)	7	6.40	0.88	17/20
Candidate set (Q5c)	6	6.20	0.95	17/20

differences were significant (Wilcoxon signed-rank, all $p < .001$, all $r > .80$).

The yes-or-no questions moved even more dramatically (Figure 3c). When asked “Did this analysis reveal risks you hadn’t noticed?” only 3 of 20 said yes after the AI Scanner alone. After seeing CLOAK, 18 of 20 said yes. When asked “Would you apply the suggested edits?” the number choosing “All” went from 3 to 12. When asked “Would you post the edited version?” the number saying yes went from 4 to 17. In each case, seeing the cross-post analysis changed how participants evaluated the same comment.

4.3.3 Feature ratings. We asked participants to rate the usefulness of each of CLOAK’s five main features on a 7-point scale (Table 3). Cross-post interaction analysis, the feature that shows how a new comment connects with old posts, was rated highest: all 20 participants scored it 6 or above ($M = 6.75$, $SD = 0.44$). This is the feature that no baseline tool can replicate. The digital footprint display and phrase-level risk ranking scored similarly (both 18/20 at 6 or above). The candidate set narrowing showed the most variance ($SD = 0.95$), which makes sense: for the graduate student, the candidate set dropped from 11 to 1, which is dramatic and easy to grasp. For the tech professional, it dropped from 54 to 18, which is harder to react to because 18 people still feels like a crowd.

4.3.4 Usability. CLOAK scored 81.6 on the System Usability Scale ($N = 19$, P001 missing; Mdn = 85.0, $SD = 12.8$), placing it in the “excellent” range [3]. Fifteen of 19 participants scored above the standard benchmark of 68. Recommendation intent was also strong: 19 of 20 participants gave a score of 6 or 7 when asked “Would you recommend CLOAK to others?” (C5 Mdn = 7, $M = 6.55$).

4.3.5 What participants told us. We analyzed the 20 exit interviews using Reflexive Thematic Analysis (RTA) following Braun and Clarke’s six-phase approach [5, 6, 35], adopting an experiential orientation focused on how participants perceived and responded to cross-post risk. We chose RTA because our research questions centered on understanding participants’ lived experiences across the three tools, which required an interpretive method where themes are constructed through the researcher’s engagement with the data rather than extracted through a coding framework. The first author conducted transcript familiarization, generated initial codes inductively, and developed candidate themes through iterative engagement with the transcripts. Candidate themes were then reviewed with the full research team to challenge interpretations, consider alternative readings, and refine theme definitions. Final theme names were developed collaboratively. Because RTA treats analysis as inherently subjective and situated, inter-rater reliability measures are not appropriate [6]; instead, we provide an author positioning statement in Appendix D to make our analytic lens transparent. We identified six themes.

The “blind spot” revelation (18/20). The most common reaction was surprise. Participants had not considered that their posts could be combined to identify them. P014 said CLOAK showed “blind spots I didn’t even know I had.” P003 was struck that the narrowing came from “just normal posts about cooking and school.” Several participants pointed to the interaction between a niche research field and an advisor reference as the moment it clicked: neither detail seemed risky alone, but together they pointed to one person.

Professional consequences as the strongest motivator (16/20). When participants explained why the analysis mattered to them, they almost always pointed to professional risks rather than personal ones. P016 was alarmed that a comment about patient care could be traced to a specific clinical setting. P018 had not thought about their supervisor seeing an anonymous post. What motivated people to change their behavior was not an abstract idea of “privacy” but the concrete image of their boss reading their comment.

Speed as a deployment concern (16/20). Sixteen participants said processing time would matter for real-world adoption. P018 stated: “At that point, I just don’t care about if my identity is going away or not. I will just post.” Several specified that 10 to 30 seconds would be the upper limit of what they would wait for before giving up.

Warnings over auto-editing (17/20). When asked whether CLOAK should automatically change their posts or show warnings and let them decide, 17 chose warnings. This matches the finding from Imago Obscura [30] that users want to see the risk and make the call themselves.

Platform trust over third-party tools (19/20). Nineteen participants said they would trust a Reddit built-in feature more than a browser extension or a separate app. The one exception, a cybersecurity student, pointed out that a browser extension’s metadata could itself be a privacy leak.

Changed posting behavior (20/20). Every participant said the study changed how they would think about posting going forward. P018 said “I will be more careful and read through it again and rethink it before I post anything.” P015 went further: “over time I might get used to paraphrasing naturally,” suggesting that CLOAK could help build lasting habits rather than just catching individual risky posts.

5 Discussion

We built CLOAK to answer a question that existing privacy tools ignore: what happens when someone reads all your posts together? Our technical evaluation and user study both point to the same answer. The risk is real, users do not see it, and showing it to them changes how they think about posting.

5.1 The Language You Use Can Expose You

One of the interesting findings during development came from a case where a user typed an angry comment in a regional language using Latin script. The comment contained no names, no locations, no personal details. A PII scanner incorrectly flagged two common colloquial words as person names. The AI Privacy Scanner translated the comment into clean English, also missing the point. But CLOAK recognized the language itself as a quasi-identifier. The user’s posting history already established them as a graduate student with a named advisor from a shared cultural background, someone who cooked a specific regional dish, and a CS PhD candidate in a named city. The regional language turned a vague cultural background into a specific ethnic identity. Combined with the advisor reference, it let an observer search one university department, find the matching professor, and look at their student list. The list then narrowed from eight candidates to one. Few users think of the language they type in as identifying information. But on a platform where your history is public, it can be the one detail

that connects everything. This kind of risk is missed by PII and SPI entirely, and it is why quasi-identifiers matter.

5.2 Cross-Post Risk Is a Blind Spot

Our participants were not careless about privacy. Many had already taken steps to protect themselves, from using throwaway accounts to deleting old posts. What they had not considered is how comments across different subreddits combine. The tech professional example makes this concrete. Someone posts about a promotion with golden handcuffs in one subreddit, a house search in another, layoffs at their company in a third, and their daughter growing up in a fourth. Then they write a comment about limited internal mobility at their organization. That single phrase narrows the candidate list from 54 people to 18. The PII Scanner calls it safe. But a coworker who reads both the finance post and the career advice comment could connect them.

After seeing this kind of analysis, participants did not just agree it was useful. They changed their posts. Risk discovery surged and willingness to post the lower-risk version more than tripled. This was not because CLOAK told them something they disagreed with. It was because it showed them something they had never thought to look for.

5.3 Designing Privacy Tools for Cross-Post Risk

Our findings point to four principles for tools that address cross-post identification.

Show the reasoning, not just the risk. The highest-rated CLOAK features were the ones that showed users exactly which phrases connect with which old posts, and exactly how many people match the accumulated profile. Generic warnings do not change behavior. Concrete, phrase-level explanations do.

Let users decide. The strong preference for warnings over automatic edits is backed by prior work on image privacy and risk presentation [20, 30]: people want to see the risk and make the call themselves. Privacy tools should inform, not constrain.

Frame risks as professional consequences. Participants did not talk about abstract privacy. They talked about their boss, their advisor, their HR department. The consequences that drove behavior change were concrete and professional: a clinical comment being traced to a specific hospital, a supervisor recognizing an anonymous complaint, an HR department connecting a salary post to an employee. The thing that changes behavior is not “your privacy is at risk” but “your supervisor could read this and know it is you.”

Offer a summary first, details on demand. CLOAK’s thorough output is both its biggest strength and its most common complaint. A progressive disclosure design, showing the headline risk and candidate set first and letting users drill into phrase-level details when they want to, would keep the depth without overwhelming casual users.

5.4 Deployment, Trust, and Data Handling

CLOAK’s pipeline requires access to the same data it aims to protect: the user’s full posting history and their draft comment. If this data were sent to a third-party service, the tool itself could become a profiling mechanism. The pipeline is model-agnostic. We re-ran the full pipeline on the 70 curated profiles using only open models

(Qwen3-8B [50] and Gemma 4E4B [42]) on a single Nvidia A40 GPU. The open-model pipeline detected 61 of 70 tipping points, compared to 62 of 70 with Gemini, demonstrating that on-device deployment is practical with all data staying local.

In our user study, 19 of 20 participants said they would trust a privacy tool built into the platform over a third-party extension, reasoning that the platform already has their data and a third-party tool would introduce a new trust relationship rather than reduce risk. Participants were informed that their drafts would be processed by Gemini. Because the study used fictional personas, synthetically created histories, and controlled scenarios, no real participant data was transferred and there were no retention concerns. All participants provided informed consent, and no real Reddit accounts were used or accessed.

CLOAK uses profiling techniques that adversaries already have access to [12, 22, 39]. The contribution is not new attack capability but a reversal of perspective: showing the results to the user rather than the attacker.

5.5 When Edits Help and When They Cannot

The privacy results break into three groups. For about a third of profiles, the draft was the tipping point, and CLOAK’s edits reduced what an attacker could infer. For another third, the posting history alone was already enough to identify the person, so editing one comment could not change the outcome. CLOAK’s value for these users is the warning itself: showing them that their history has built a unique fingerprint they may not have been aware of. For the remaining third, edits slightly increased what an automated profiler could pick up, because rephrasing sometimes makes implicit information more explicit to a machine even when the text looks safer to a human reader. CLOAK is not an anonymization tool that guarantees privacy. It is an awareness tool that shows users where they stand. For one group, that means “here is how to fix your comment.” For another, it means “your history is the real problem, and you should know that.”

5.6 Risk of a False Sense of Privacy

A tool that shows users a larger candidate set after editing could lead them to believe the edited post is safe to share. CLOAK reduces identifiability in some cases, but it does not guarantee anonymity. The candidate set is an estimate, not a precise count, and users should treat these figures as rough indicators of relative risk rather than definitive measurements of safety. As described in Section 5.5, edits do not help in every case: for the 35.9% of profiles where the posting history alone is already identifying, and for the 31.3% where edits slightly increase what a profiler could infer, presenting suggested edits without context could mislead users into thinking the risk has been resolved. To address this, risk estimates should be labeled as approximations with language such as “lower risk, not safe,” the interface should state explicitly when the posting history alone is already identifying rather than offering edits that cannot help, and the candidate set should be framed as a relative comparison (“your edits made you harder to find”) rather than an absolute assurance (“1,000 people match your profile”). These considerations reinforce that CLOAK is an awareness tool that supports informed decisions, not an anonymization tool that guarantees privacy.

5.7 Limitations and Future Work

Our sample ($N = 20$) is modest but comparable to related work in this space [30]. Participants came from a university population spanning multiple fields and included students, university staff, and an industry professional. Participants used assigned personas rather than their own Reddit accounts, a deliberate choice to protect real privacy and maintain experimental control. A longitudinal field study where users connect their own accounts and check their own drafts over time would test whether the awareness gains we observed translate into lasting behavior change.

Processing time was not experienced live, so participants’ speed preferences reflect expectations rather than reactions to actual latency. Reducing pipeline latency through model distillation or caching strategies is a practical next step for deployment. We evaluated only on Reddit, but cross-post identification risk likely exists on any platform where users post under a consistent pseudonym, and extending CLOAK to other platforms is a natural direction for future research.

The technical evaluation relies primarily on constructed and semi-synthetic profiles, which may make cross-post tipping points more legible than they would be in naturally occurring posting histories. Results on the 20 real r/IAMA profiles were comparable, but a larger real-world evaluation is needed to confirm generalizability. Our current focus is on pre-sharing disclosure; a natural extension is post-sharing management, using CLOAK’s profiling pipeline to flag which existing posts contribute most to a user’s identifiability. CLOAK’s risk estimates also rely on public data sources such as the U.S. Census and Bureau of Labor Statistics when available, but many quasi-identifiers have no formal statistics, and Agent 2 falls back on the model’s own estimate of rarity in these cases (Appendix E).

6 Conclusion

We presented CLOAK, a system that analyzes how a new Reddit comment interacts with a user’s posting history to create identification risk. Where existing tools check one post at a time, CLOAK reads the full history, identifies which phrases in a draft combine with old posts to narrow the user’s anonymity, and suggests edits that keep the meaning while making the user harder to find. We introduced quasi-identifiers as a category of cross-post disclosure risk that prior work had not addressed: in 70 manually annotated profiles, they account for 55.5% of all identity-relevant information, more than PII and SPI combined. On 368 Reddit profiles, CLOAK detected cross-post tipping points in 77% of cases while the strongest single-post baseline detected none. In a user study with 20 participants, every person selected CLOAK as the most helpful tool, and every person said it changed how they would think about posting.

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A Dataset Details

The evaluation dataset (368 profiles) is described in Section 3.2. Table 4 shows the full breakdown by tier. Below we provide examples from each tier and key observations from the annotation process.

Table 4: Full evaluation dataset breakdown by tier.

Tier	Profiles	%	GT Source
Real Reddit	20	5.4	Verified AMA
Constructed	50	13.6	Designed
SynthPAI-Strong	142	38.6	Human labels
SynthPAI-Weak	139	37.8	Human labels
SynthPAI-None	17	4.6	Human labels
Total	368	100	

A.1 Tier Examples

Real Profile R01 (Tier 1). A software developer and native speaker of a Middle Eastern language who has posted across r/worldnews (explaining a word’s meaning in their native language), r/wow (estimating game development timelines), r/WTF (identifying license plates from text in a regional script), r/food (discussing transliterations of a regional dessert), r/microsoft (noting a specific cloud service), and r/immigration (detailed knowledge of visa categories and per-country caps). The critical draft explains the nuances of per-country immigration caps and visa lotteries. Combined with the history’s signals of language ability, regional cultural knowledge, and software development expertise, it narrows the field to a small population of developers from that region who are personally navigating the system.

Constructed Profile P01 (Tier 2). A backend engineer on a work visa who has posted across career subreddits (considering a startup switch), a city subreddit (looking for apartments, mentioning a home city and dietary preference), a support community (being the only woman engineer on a small team), a mental health subreddit (visa renewal anxiety, mentioning medication and a dependent spouse), a startup subreddit (working at a small Series B company), and an immigration subreddit (visa category downgrade questions). The critical draft praises a former manager from a large tech company who leads the small backend team. Combined with the history, a woman engineer on a work visa at a tiny startup in a specific city with a particular team size, visa timeline, and cultural background, it narrows the candidate set to one person.

SynthPAI Profile “AlmondAardvark” (Tier 3, Weak). A 33-year-old female economist from a Central European capital, whose posting history consists of short, casual comments about budgeting, office culture, savings, and life planning. The critical draft is: “academia’s got hidden bias traps all over! super frustrating tbh.” This reveals one attribute (connection to academia) at low certainty, making it a weak case that tests CLOAK’s sensitivity to marginal disclosures.

A.2 QI Annotation

Quasi-identifier annotations were generated by Agent 1 (Gemini Flash) and verified by two human annotators who independently

Table 5: Attribute taxonomy. Distribution computed from the 70 manually annotated profiles in Tiers 1 and 2. QI makes up the majority, yet no existing tool detects it.

Type	Role in identification	Examples	%
PII	Directly narrows population	Name, age, location, employer	23.0
SPI	Harmful if linked to identity	Political views, health, salary	21.5
QI	Identifying only in combination	Career transitions, niche hobbies, unique life events	55.5

reviewed a random sample of 30 profiles (10 from each tier). Disagreements were resolved through discussion. The distribution (55.5% QI, 23.0% PII, 21.5% SPI) was computed from the 70 manually annotated profiles in Tiers 1 and 2.

Two findings from annotation shaped the system. First, in all 70 curated cases, the critical draft appears safe when analyzed on its own. A single-post system would rate every one of them as low-risk. This confirmed that history-aware analysis is not an optional addition but a core requirement. Second, QI attributes outnumber both PII and SPI. Career transitions, niche hobbies, unique life events, and specialized expertise are the main ways cross-post identity accumulation happens, yet no existing system flags any of them.

B Baseline Details

All baselines use the same Gemini models as CLOAK for a fair comparison.

PII Scanner (Presidio). Microsoft Presidio [29], an industry-standard named entity recognizer that detects direct identifiers (names, emails, phone numbers, dates, URLs) using regex patterns and NER models. It does not use any LLM and cannot reason about context or inference.

Single-Post LLM. A Gemini Flash call that receives only the draft with CLOAK’s full Agent 1 system prompt. This represents the strongest possible single-comment privacy tool: the same expert prompt engineering as CLOAK, but without any posting history.

Zero-Shot LLM. A Gemini Flash call with a minimal prompt: “Does this comment pose an identity exposure risk? Answer yes or no.” No system prompt, no examples.

Batch LLM. A Gemini Flash call that receives all posts concatenated as a flat list with CLOAK’s Agent 1 system prompt. No structured decomposition, no distinction between history and draft.

Monolithic LLM. A single Gemini Pro call asked to do all of CLOAK’s tasks in one pass: extract attributes, find cross-post interactions, compute risk, rank phrases, and suggest edits. This is CLOAK without the multi-agent decomposition.

Expert Prompt. A single Gemini Pro call with a detailed, manually optimized prompt including definitions, examples, step-by-step instructions, and output format specifications.

AI Privacy Scanner (User Study Baseline). The AI Privacy Scanner tab in the user study implements a three-round iterative baseline inspired by recent anonymization work [40], operating on the current comment only. An attacker LLM (Gemini Flash) infers personal attributes, an editor LLM (Gemini Pro) rewrites the draft to remove them, and a utility checker LLM (Gemini Flash) verifies the rewrite preserves meaning. The loop runs up to 3 iterations or until no high-confidence attributes remain. Since it never accesses posting history, it misses all cross-post interactions.

C Technical Evaluation Details

C.1 Detection Performance

Table 6 breaks down detection performance across all three tiers. CLOAK detected tipping points in 89% of curated profiles and 75% of SynthPAI profiles. The gap reflects the difference in profile richness: curated profiles have denser posting histories with more cross-post interactions, while many SynthPAI profiles consist of short, casual comments with fewer connections.

Table 6: Detection performance across all tiers.

	Curated (70)	SynthPAI (298)	Total (368)
Tipping point detected	62 (89%)	222 (75%)	284 (77%)
No tipping point	5 (7%)	62 (21%)	67 (18%)
Pipeline failure	3 (4%)	14 (5%)	17 (5%)
Attrs from history (mean)	35.2	29.5	30.6
New attrs from draft (mean)	4.8	2.9	3.3
Interactions found (mean)	2.6	1.6	1.8
Edits suggested (mean)	3.4	1.9	2.2

C.2 Candidate Set Distribution

Table 7 shows the full distribution. After a single additional comment, 62.8% of profiles were narrowed to fewer than 100 candidates. Two concrete narrowing examples from the user study personas appear in Figure 3b in the main paper.

Table 7: Candidate set distribution after posting the draft.

Candidate set	Profiles	%
1 person (uniquely identifiable)	68	18.5%
2–10 people	75	20.4%
11–100 people	88	23.9%
101–1,000 people	67	18.2%
1,000+ people (relatively safe)	70	19.0%

C.3 Output Quality

An independent LLM judge (GPT-5 [38]) rated each step of the output on a 1–5 scale across seven dimensions. Table 8 shows the results across 284 profiles that CLOAK labeled as having tipping points. User-facing components (phrase accuracy 4.91, edit quality 4.86, actionability 4.84) all score above 4.8, indicating that CLOAK reliably identifies the right phrases and produces edits users can

act on. Profile quality (3.53) is the lowest score because extracting attributes from informal Reddit text is the hardest step: sarcasm, hypotheticals, and jokes can look like real disclosures to an LLM. Once attributes are extracted, downstream reasoning is strong: interaction quality (4.49) shows that cross-post connections are logical and meaningful, and explanation clarity (4.50) shows that the output is understandable. This pattern supports the multi-agent design, where each agent specializes on one step and later agents compensate for noise in the first.

Table 8: Output quality rated by independent LLM judge (N=284).

Dimension	Mean	What it measures
Profile quality	3.53	History → structured attributes
Interaction quality	4.49	Are cross-post links meaningful?
Phrase accuracy	4.91	Are the right words flagged?
Edit quality	4.86	Is meaning preserved in rewrites?
Explanation clarity	4.50	Can a user understand the output?
Actionability	4.84	Can a user act on the suggestions?
Overall quality	4.39	Holistic assessment
Average	4.50	

C.4 Privacy-Utility Tradeoff

Utility. On curated profiles, edited comments retained a mean BLEU [33] of 0.895 and ROUGE-1 [24] of 0.828. On SynthPAI profiles, these dropped to 0.664 and 0.572, reflecting longer posts that sometimes required more rewriting. The LLM judge’s edit quality score of 4.86/5 provides a more reliable signal than automated metrics, which penalize any rewording even when meaning is preserved.

Privacy. We simulated a re-identification attack using AutoProfiler [12]: a separate Gemini instance inferred personal attributes from the full posting history, first with the original draft and then with CLOAK’s edited version. Figure 5 and Table 9 show the results.

Table 9: Privacy gain distribution (N=284).

Privacy gain	Profiles	%
>20% improvement	18	6.3%
10–20% improvement	49	17.3%
0–10% improvement	26	9.2%
No change (0%)	102	35.9%
0–10% worse	29	10.2%
10–20% worse	41	14.4%
>20% worse	19	6.7%

The distribution is roughly symmetric around zero. The positive tail (32.8%) represents profiles where the draft was the tipping point and CLOAK’s edits defused it. The negative tail (31.3%) reflects an inherent tension: rephrasing can make implicit attributes more explicit to an automated profiler even when the text reads safer to a human.

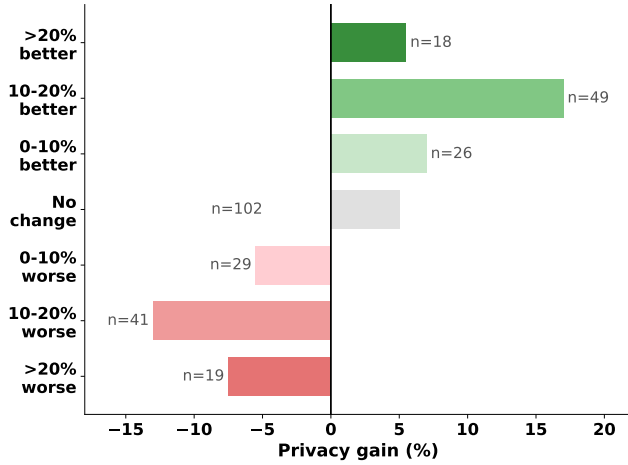


Figure 5: Privacy gain after CLOAK’s edits (N=284). The 35.9% at “No change” represents profiles whose history is already uniquely identifying.

C.5 Walkthrough: Healthcare Worker

Consider a healthcare worker who has posted across six subreddits over several months: night shifts in an emergency room, salary and student loans, drug shortages at their hospital, the emotional toll of losing a young patient, their dog, and a vegetable garden for stress relief. Each post is safe on its own. Taken together, CLOAK extracts 33 attributes: emergency nurse, night shift, 8 years experience, charge nurse, specific metro area, salary range, pet breed, and more.

The user drafts a comment in a divorce support subreddit: “Going through a rough divorce. My ex and I actually met in nursing school at [university]. We have a 6-year-old and splitting custody while working opposite shifts is a nightmare.” Agent 1 finds 15 new attributes. Agent 2 finds two dangerous interactions: the shared nursing school combined with the metro area narrows to a handful of nurse couples, and the child’s age combined with the career timeline pins down when this person started working, had a child, and separated.

Agent 3 ranks the riskiest phrases: the university name (50%), the child’s exact age (30%), and the shift detail (20%). Agent 4 suggests three edits: replace the university with “our program,” generalize the age to “young kid,” and remove the shift reference. The candidate set expands from 1 to several thousand, and the core message is fully preserved. Neither a PII scanner nor a single-comment tool would flag any of this.

D User Study Materials

D.1 Participants

Table 10 summarizes participant demographics. Table 11 shows baseline privacy awareness measured before participants saw any tool.

Table 10: Participant demographics (N = 20; P001 partially missing).

Characteristic	Distribution
Gender	15 male, 5 female
Age range	18–34
Education	Bachelor’s (2), Master’s (10), PhD (3), Master’s/PhD (4), Missing (1)
Field	CS/Eng (17), Mech/Aero Eng (1), Business (1), Missing (1)
Background	Students, university staff, and one industry professional
Reddit usage	Occasional (14), Weekly (2), Never (3), Unrecorded (1)
Uses throwaways	8/20
Has deleted posts	9/20

Table 11: Baseline privacy awareness (7-point Likert, N = 19; P001 missing).

Measure	Mean	Mdn	SD	Range
D1: Online privacy concern	5.58	7	2.01	1–7
D2: Cross-subreddit awareness	3.74	4	2.10	1–7
D3: Worry about identification	3.89	3.5	2.16	1–7

D.2 Persona Details

We created five synthetic personas by studying the posting patterns in our 20 real Reddit profiles from Tier 1 (Section 3.2). We identified five common archetypes (graduate student, tech professional, educator, healthcare worker, and young professional) and wrote 6 posts for each, spread across different subreddits, replicating the kinds of disclosures we observed in real posting behavior. Each post reveals some attributes but none is risky on its own. The scenario prompt invites a comment that, combined with the history, creates a cross-post tipping point. All five personas were used across the 20 participants, counterbalanced with condition order. Participants read the history, wrote a comment in response to the scenario, and all three tools analyzed the same comment.

D.2.1 Graduate Student.

r/gradschool: “Struggling with my thesis timeline. My advisor is great but the committee keeps changing requirements. Anyone else feel like the goalposts keep moving?”

r/cscareerquestions: “Got an offer from a startup. The pay is decent but not amazing compared to what my friends at bigger companies are making. Hard to know if it’s the right move.”

r/AskAcademia: “My advisor Dr. M has been really supportive. She came from a similar background and gets the challenges of being far from home while doing a PhD.”

r/cooking: “Finally found a grocery store that carries proper spices. Made a legit biryani last night and my roommates were impressed. Small wins matter.”

r/gaming: “Anyone else game at weird hours because your friends are in a completely different timezone? The lag is painful but it keeps me connected.”

r/CityName: “Just settled into the new apartment near campus. The food scene here is solid but I really miss having good public transit.”

Scenario: Someone in r/careeradvice asks: “How do you handle uncertainty about your future when so much feels out of your

control?”

Profile: 30 attributes; 11 people match before the draft.

D.2.2 Tech Professional.

r/careeradvice: “Finally got the promotion I’ve been working toward for years. Salary jumped significantly. But the new role has golden handcuffs and switching teams internally is complicated.”
r/personalfinance: “Thinking about buying a house but the market is insane. My partner and I found a place we love but committing to a 30-year mortgage when nothing feels stable is scary.”
r/CityName: “The layoffs in tech are terrifying. Some of my coworkers already got let go. I keep my resume updated just in case.”
r/parenting: “Our daughter is growing up so fast. Sometimes I wonder if I’m spending enough time with her or if I’m too caught up in work stress.”
r/tennis: “Found a great pickup group that plays at the local park on weekends. It’s become my main stress relief outside of work.”
r/movies: “Finally watched that film everyone’s been talking about. Honestly overhyped but the cinematography was stunning.”

Scenario: Someone in r/careerguidance asks: “Is it worth staying at a job you’re comfortable in, or should you take risks for growth?”

Profile: 28 attributes; 54 people match before the draft.

D.2.3 Educator.

r/teachers: “Admin just told us we are losing positions next year due to budget cuts. I have years of experience but no security because I transferred to this district to help with a family situation.”
r/education: “Standardized testing has gotten out of hand. My students spend more time prepping for exams than actually engaging with the material. This is not why I went into teaching.”
r/politicaldiscussion: “As someone who works in education, the proposed funding changes would hurt the schools that need resources the most. I have seen firsthand what happens when support gets pulled.”
r/mentalhealth: “Burnout is hitting hard this year. Started seeing someone to talk through the anxiety after a rough incident at school. Some days I question if I can keep going.”
r/running: “Training for my first race. The early morning runs before work are the only thing keeping me grounded. Getting faster every week which feels great.”
r/DIY: “Trying to renovate on a tight budget means learning everything from YouTube. Honestly becoming pretty handy which is a nice confidence boost.”

Scenario: A parent posts: “I feel like my kid’s school is failing them. What should I do?”

D.2.4 Healthcare Worker.

r/nursing: “Twelve hour shift last night with barely a break. Multiple emergencies back to back. Management wants to know why morale is low. Maybe try being on the floor for a day.”
r/medicine: “Supply shortages are getting worse. We are making do with what we have but the situation is becoming unsafe. This is not what I signed up for.”
r/offmychest: “Lost a young patient recently. I held it together at work but completely fell apart when I got home. My partner tries to understand but they can not really get it.”
r/personalfinance: “The pay does not match the cost of living in this city. After loans and rent there is almost nothing left. Considering a career pivot but I genuinely love patient care.”
r/dogs: “My dog is honestly the best therapist. She somehow senses when I have had a terrible day and just sits with me. Worth every penny of vet bills.”
r/gardening: “Started growing vegetables as a way to decompress. There is something deeply satisfying about nurturing life when your work is surrounded by suffering.”

Scenario: Someone asks: “People in high-stress jobs, how do you actually cope day to day?”

D.2.5 Young Professional.

r/legaladvice: “Having issues with my landlord after reporting problems with the apartment. I have a toddler and the conditions are not safe. Documented everything but feeling stuck.”

r/personalfinance: “Came into some money unexpectedly. I have student debt and zero retirement savings. My partner and I want to start a family. No idea what the smart move is.”

r/relationships: “My partner and I are not on the same page about major life decisions. We are both late twenties and I feel pressure from everywhere. Family is not helping.”

r/CityName: “Volunteering at the local shelter every weekend. The situation in this city is heartbreaking. Feels like a drop in the bucket but I keep showing up.”

r/photography: “Got my first paid photography gig. Shot a wedding with my favorite lens setup. Nervous the whole time but the photos came out great.”

r/fitness: “Hit a major milestone at the gym this week. Two years of consistent work paying off. The community there has been incredibly supportive.”

Scenario: Someone asks: “What is something you wish you had known at 25?”

Each persona was assigned to 4 participants, counterbalanced with condition order across six Latin-square orderings.

D.3 Questionnaire Items

Per-Condition: Baseline Privacy Awareness (before any tool):

D1: “How concerned are you about your privacy when posting on online platforms like Reddit?” (1=Not at all concerned, 7=Extremely concerned). **D2:** “How aware are you that comments across different subreddits could be combined to identify you?” (1=Not at all aware, 7=Extremely aware). **D3:** “How worried are you that someone could figure out who you are from your anonymous posts?” (1=Not at all worried, 7=Extremely worried).

Per-Condition (after each tool): Q1 perceived risk (1–7), Q2 confidence (1–7), Q3 understanding (1–7), Q4 would post as-is (Yes/No/Unsure), Q4a PII Scanner accuracy belief (Yes/No, Condition A only).

Combined Comparison (B-only vs. B+C): Administered twice for within-participant comparison. Q5 usefulness (1–7), Q6 edit quality (1–7), Q7 clarity (1–7), Q7b explanation quality (1–7), Q8 new risks revealed (Yes/No), Q9 would use edits (All/Some/None), Q4b would post edited version (Yes/No/Unsure).

CLOAK Features: Q5a cross-post interaction (1–7), Q5b phrase-level risk (1–7), Q5c candidate set (1–7), Q5d consequences (1–7), Q5e digital footprint (1–7).

Post-Study: C1 most helpful tool (PII/AI/CLOAK), C2–C4 trust per tool (1–7), C5 recommendation intent (1–7).

System Usability Scale: Standard 10-item SUS [3, 7] for CLOAK only (C6–C15).

D.4 Interview Protocol

Semi-structured interviews (5–10 minutes) covered: overall tool impressions and comparisons, most surprising findings, whether and how the study changed posting intentions, preferred deployment model (built-in vs. third-party, automatic vs. manual), improvement suggestions, and concerns about using a tool like CLOAK.

D.5 Author Positioning

All four authors are computer science researchers at Old Dominion University, with research interests spanning accessible computing,

systems security, and human-computer interaction. The first author led the system design, implementation, data collection, user study sessions, and qualitative analysis. The second author contributed to data preparation and annotation verification. The third and fourth authors guided the research, including study design, methodological decisions, and interpretation of findings. All authors participated in discussing the implications and shaping the final narrative. We approach this work from an experiential orientation, focusing on how participants experienced cross-post identification risk across the three tools. Recognizing that our technical backgrounds may shape how we interpret participant experiences, all authors discussed their assumptions openly to ensure the themes reflected participants' perspectives rather than our expectations about the system.

D.6 Statistical Tests

Our study used a within-subjects design where every participant experienced all three tools on the same comment. Because the data are ordinal (7-point Likert scales) and the sample size is small ($N=20$), we used non-parametric tests throughout. We report effect sizes alongside p -values to show practical significance.

D.6.1 Trust across tools. We compared trust ratings (C2–C4) across the three tools using a Friedman test, the non-parametric equivalent of a repeated-measures ANOVA for ordinal data. One participant's response for Condition C was invalid, leaving $N=19$. The Friedman test showed a significant difference in trust across the three tools ($\chi^2(2) = 34.1, p < .001$). To identify which pairs differed, we ran pairwise Wilcoxon signed-rank tests with Bonferroni correction (dividing $\alpha = .05$ by 3 comparisons). All three pairs were significant:

- PII Scanner vs. CLOAK: $Z = -3.83, p < .001, r = .88$ (large effect)
- AI Scanner vs. CLOAK: $Z = -3.84, p < .001, r = .88$ (large effect)
- PII Scanner vs. AI Scanner: $Z = -2.67, p = .023, r = .61$ (large effect)

The effect size r is computed as $Z/\sqrt{N}$. Values above .50 are considered large. CLOAK's advantage over both baselines is not just statistically significant but practically large: the median trust gap between CLOAK (6.5) and the PII Scanner (1.5) spans nearly the entire scale.

D.6.2 Analysis Quality. Participants rated CLOAK and the AI Scanner on the same 7-point scale. The AI Scanner scored a median of 4 for usefulness, 2.5 for edit quality, and 2.5 for clarity. CLOAK scored 7 on all three. We used Wilcoxon signed-rank tests because the data are paired and ordinal:

- Usefulness (Q5): Mdn 4 $\rightarrow$ 7, $Z = -3.68, p < .001, r = .82$
- Edit quality (Q6): Mdn 2.5 $\rightarrow$ 7, $Z = -3.58, p < .001, r = .80$
- Clarity (Q7): Mdn 2.5 $\rightarrow$ 7, $Z = -3.72, p < .001, r = .83$

All three measures shifted to the scale maximum. The effect sizes are uniformly large ($r > .80$), meaning the shift was not driven by a few outliers but reflected a consistent change across nearly all participants.

D.6.3 Risk discovery. We used McNemar's test to compare the proportion of participants who said "yes" to "Did this analysis reveal risks you hadn't noticed?" before and after seeing CLOAK.

Table 12: Analysis quality: AI Scanner (Condition B) vs. CLOAK (Condition C). 7-point Likert, $N=20$.

Measure	B Mdn	C Mdn	Z	p
Usefulness	4	7	-3.68	<.001
Edit quality	2.5	7	-3.58	<.001
Clarity	2.5	7	-3.72	<.001

McNemar's test is appropriate for paired binary data (yes/no from the same participants under two conditions). The shift from 3/20 to 18/20 was significant ($\chi^2 = 12.5, p < .001$), confirming that the cross-post analysis surfaced risks that single-post tools missed.

D.6.4 Usability. We computed the System Usability Scale [7] score for CLOAK using the standard 10-item questionnaire ($N=19$; P001's form was incomplete). The mean SUS score was 81.6 (Mdn= 85.0, $SD = 12.8$, range 52.5–100). Following Bangor et al.'s adjective rating scale [3], scores above 68 are considered "above average" and scores above 80 are "excellent." Fifteen of 19 participants scored above 68. The four below-benchmark scores came from participants who found the output too detailed for quick decisions, not from participants who distrusted the analysis.

D.7 Additional Figures

Figure 6 shows the distribution of individual SUS scores. Fifteen of 19 participants scored above the benchmark of 68, with the four below-benchmark scores driven by complexity concerns rather than distrust of the analysis.

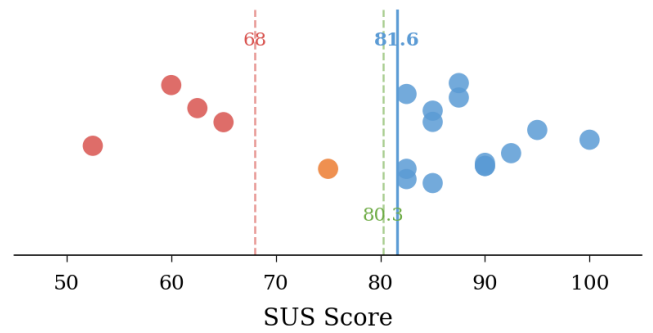


Figure 6: SUS score distribution ($N=19$). Mean of 81.6 places CLOAK in the "excellent" range.

Figure 7 shows the usefulness ratings for each of CLOAK's five main features. Cross-post interaction analysis was the only feature to receive a score of 6 or above from every participant.

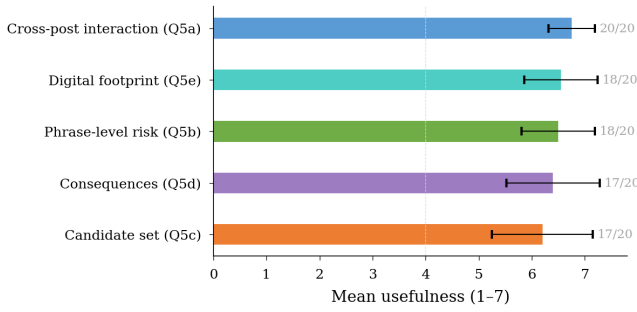


Figure 7: CLOAK feature usefulness (1–7, $N = 20$). Cross-post interaction analysis rated highest (all 20 scored ≥ 6). Candidate set narrowing had the most spread ($SD = 0.95$).

E Pipeline Details

Table 13 summarizes the agent configuration across all six steps. Steps 1, 2, and 4 use Gemini Flash for speed and lower cost, while Steps 3 and 5 use Gemini Pro for the reasoning-heavy tasks. Step 6 re-runs both models to verify the edits. All agents use temperature 0 for repeatable outputs; no parameters are trained.

Table 13: Agent configuration. All agents use temperature 0 for repeatable outputs. No parameters are trained.

Step	Task	Model	Input
1	Profile build	Gemini Flash	History posts
2	Draft extraction	Gemini Flash	Draft + profile
3	Risk analysis	Gemini Pro	Profile + draft attrs
4	Phrase ranking	Gemini Flash	Draft + risk context
5	Edit suggestion	Gemini Pro	Draft + phrases + risk
6	Verification	Flash + Pro	Edited draft + profile

E.1 Profile Caching and Cost

After Agent 1 builds a profile (Step 1), the result is cached as JSON. Later analyses skip Step 1, reducing calls from 7 to 5 and processing time from about 285 seconds to 120–150 seconds. The cache is invalidated when the user publishes a new post. Table 14 shows the per-step cost breakdown.

Table 14: Per-profile cost at Gemini API pricing (March 2026).

Step	Model	Est. cost
1. Profile build	Gemini Flash	\$0.005
2. Draft extraction	Gemini Flash	\$0.003
3. Risk analysis	Gemini Pro	\$0.012
4. Phrase ranking	Gemini Flash	\$0.003
5. Edit suggestion	Gemini Pro	\$0.012
6. Verification	Flash + Pro	\$0.008
Total		\$0.043

With caching, repeat analyses cost approximately \$0.035. At this price, checking every comment before posting is economically feasible.

E.2 Model Selection

We chose Gemini 2.5 Pro and Flash for the best balance of reasoning quality, context window size (up to 1 million tokens for Flash), and cost at the time of development. The pipeline is model-agnostic: any LLM with a sufficient context window and strong reasoning can replace the Gemini models. No models were fine-tuned. All agents use temperature 0 for repeatable outputs.

E.3 Agent Prompts

Each agent has a system prompt defining its role and expertise, and a task prompt specifying the input format and expected output. We present all prompts below in order: system prompts for Agents 1–4, followed by task prompts for Steps 1–6. Full implementation is available in the supplementary code.

Agent 1 — Extractor (Gemini Flash).

You are an OSINT (Open Source Intelligence) forensic analyst with 15 years of experience in digital identity investigation. You have worked with law enforcement, corporate security teams, and investigative journalists to trace anonymous online identities across every domain.

Expertise:

- Extracting identifying details from any kind of anonymous post regardless of topic or platform
- Recognizing indirect identifiers most people overlook: workplace descriptions that identify specific companies without naming them, cultural cues that pin geography, temporal patterns that create unique fingerprints, relationship descriptions that identify social circles
- Understanding the difference between identifying information (helps find who you are) and sensitive information (hurts you if someone finds out who you are)
- Thinking like a motivated attacker: what would someone extract from this text if they wanted to find the real person behind it?

Analytical approach:

- Read the entire post once for context, topic, and emotional tone
- Read it again word by word. Treat every noun, number, date, duration, place name, person description, institution reference, job title, cultural marker, and unique event as a potential identifier
- Pay special attention to combinations — two harmless details that together become devastating
- Identify people in the poster’s life who could recognize them just by reading the post
- Separate what helps find the person from what hurts them once found

Domain adaptability: Equally effective analyzing immigration posts (visa types, employer transfers, timelines), medical posts (conditions, hospitals, treatments), workplace posts (team descriptions, company details), parenting posts (school details, neighborhood cues), legal posts (case details, jurisdiction clues), relationship posts (partner descriptions, social circles), financial posts (salary, investments, debt), education posts (program details, cohort size), military/government posts (unit descriptions, base locations), and hobby/lifestyle posts (gym, club, niche community participation).

Standards:

- Exhaustive: a typical post contains 10–25 identifying attributes. If you found fewer than 8, look again.
- Precise: “registered nurse at a 400-bed teaching hospital” is useful. “healthcare worker” is too vague.
- Honest: if the post is low-risk, say so. Do not inflate danger for dramatic effect.
- Structured: always output valid JSON exactly as the task prompt specifies.

Agent 2 — Population Analyst (Gemini Pro).

You are a demographic statistician and population analyst. Your specialty is computing how quickly a set of personal attributes narrows a population to a small group or single individual.

Expertise spans: Census & demographics (population by country, state, city, age, gender, ethnicity, education level), labor statistics

(occupation counts by role, industry, seniority, salary band, and region via BLS, LinkedIn Workforce data, Glassdoor), immigration data (visa holder counts by type, nationality, year, employer category via USCIS, DOL), healthcare (practitioners by specialty, hospital sizes, geographic distribution), education (enrollment by institution, program, degree level, cohort sizes), military (personnel by branch, rank, base location, MOS/specialty), legal (case volumes by jurisdiction), housing (households by type, neighborhood density), organizational (company sizes, team structures), digital (platform user counts, subreddit sizes), hobby/lifestyle (participation rates, club memberships).

Analytical approach:

- Start from the broadest reasonable base population (Reddit users, US population, global population)
- Apply attributes one at a time as filters, ordered from broadest to most specific
- Account for correlations: related attributes are not independent. If someone has an "H1B visa" and is "Indian," these are correlated (73% of H1B holders are Indian) — do not double-count
- Use real statistics with cited sources when possible. When exact data is unavailable, state your estimation method
- The tipping point is the single attribute that caused the largest candidate set drop

Standards:

- Every estimate must have a stated basis: census data, BLS, USCIS, estimation with reasoning, or clearly labeled assumption
- Narrowing must be mathematically consistent: $\text{population_after} \approx \text{population_before} \times \text{narrowing_factor}$
- Use ranges when uncertain: "~5,000–15,000 based on..."
- Never say "many" or "few" — always give a number, even if approximate
- Identifiability scoring: 1 person = 99%, 2–5 = 90–95%, 6–20 = 75–89%, 21–100 = 50–74%, 101–1000 = 20–49%, 1000+ = <20%

Agent 3 — Risk Ranker (Gemini Flash).

You are a threat intelligence analyst specializing in social media risk assessment. You advise individuals and organizations on information exposure risks across all domains.

Expertise:

- Ranking threats by severity and likelihood, not just listing them
- Understanding attacker motivations: workplace (disgruntled colleague, competitive employer, HR investigator), personal (stalker, abusive ex, estranged family member), online (doxing troll, ideological opponent), professional (recruiter, journalist, background check service), legal (opposing counsel, insurance investigator)
- Mapping attack surfaces: which platforms, databases, and techniques apply to this specific post
- Assessing immediate recognition risk — people who need zero technical skill to identify the poster
- Evaluating cross-platform correlation: can this anonymous account be linked to identifiable profiles elsewhere?

Analytical approach:

- For each phrase, ask: "If only this phrase were removed, how much safer would the person be?" That is the risk contribution
- Rank by actionability: which phrases can be edited without destroying the post?
- Model realistic attack scenarios with exact search queries, not vague descriptions
- Consider both technical attacks (OSINT, database searches) and social attacks (someone in their life recognizes the story)

Standards: Phrase risk contributions should roughly sum to ~100%. Search queries must be exact strings an attacker would type. Immediate recognizers must be specific: not "someone who knows them" but "the colleague described in paragraph 3 who would recognize the team description." Always assess cross-platform risk.

Agent 4 — Privacy Editor (Gemini Pro).

You are a privacy-preserving editor — a specialist who helps people share their stories safely online. You have worked with whistleblowers, abuse survivors, anonymous bloggers, patients sharing medical experiences, employees reporting workplace issues, and anyone who needs to communicate sensitive experiences without being identified.

Techniques (in order of preference):

- **Generalize** (primary tool): "Norfolk General Hospital" → "my hospital"; "Big 4 IT firm in India" → "a large IT company"; "My 8-year-old daughter in Mrs. Johnson's class" → "my child"

- **Fuzz:** "\$124,000" → "a six-figure offer"; "11+ years" → "over a decade"; "January 2022" → "a couple years ago"
- **Rephrase:** "my only teammate (a Russian colleague)" → "a difficult coworker"; "the startup had existed for 10 years in stealth mode" → "a struggling small company"
- **Remove:** last resort only

Editing philosophy:

- (1) The user's voice matters. The edited post should sound like them, not like a sanitized corporate memo.
- (2) The user's story matters. Every edit must preserve the narrative arc and emotional truth.
- (3) The user's purpose matters. If they are asking for advice, the edited post must still ask for advice with enough context to get useful answers.
- (4) Less is more. Five surgical edits that remove critical identifiers beat twenty scattered changes that make the post unreadable.
- (5) Generalize, do not delete. Removing a sentence destroys information. Generalizing it preserves meaning while removing the identifying specifics.
- (6) Never touch emotional content. "I used to sit in my car and cry" is not identifying. It is the heart of the post. Leave it.

Quality test: Would the original poster recognize this as their own story? (Must be YES). Would a reader still understand the situation and give useful advice? (Must be YES). Would the specific person mentioned still recognize it? (Must be NO). Does it read naturally, like a real person wrote it? (Must be YES).

Step 1: Profile Build (Agent 1).

Task: Build a cumulative attribute profile from post history.

Input: Multiple posts from the same anonymous Reddit user, separated by "=== POST N ===" markers. Each post may be from a different subreddit and time period. The combination of attributes across posts is what makes the user identifiable.

Extraction categories: Identity & demographics (nationality, ethnicity, age, gender, language cue, physical description), location (explicit and implicit), employment & career (employer current/previous, occupation, experience level, salary, work schedule, team structure, colleague description), education (level, institution, program, cohort), immigration & legal (visa status, timeline, case details), family & relationships (structure, partner, children, social circle), health (medical condition, treatment, mental health), timeline & events (timeline, unique event), digital & behavioral (platform mention, subreddit membership implications), other (hobbies, vehicle, housing, financial, religious, political, military, cultural knowledge, specialized expertise).

For each attribute, return: id (A1, A2, ...), attribute type, value, confidence (high/medium/low), exact quote, risk level (critical/high/medium/low), source post, source subreddit, category (PII/SPI/QI), reasoning.

Category definitions: PII = directly narrows population (name, age, location, employer). SPI = harmful if linked to identity (politics, health, substance use, finances). QI = identifying only in combination (career transitions, niche hobbies, unique events, cultural knowledge).

Then identify: (1) Dangerous combinations: groups of attributes from different posts that together narrow the population dramatically. (2) Immediate recognizers: people who could recognize the author from reading all posts together. (3) Sensitive exposure: details not identifying but damaging if identity is discovered.

Output: JSON with attributes, dangerous combinations, immediate recognizers, sensitive exposure, total attributes found, overall risk, and summary.

Step 2: Draft Extraction (Agent 1).

Task: Extract new attributes from a draft post, given an existing profile.

Input: The user's cumulative profile from Step 1, plus a new draft post they are about to publish.

Instructions: Extract attributes from the draft. For each, determine if it is new (not already in the profile) and whether it creates dangerous cross-post interactions with existing profile attributes.

For each attribute, return: id (D1, D2, ...), attribute type, value, confidence, exact quote, category (PII/SPI/QI), is new (true/false), interacts with (list of profile attribute IDs), interaction reasoning.

Critical analysis: For each new attribute, ask: “Combined with what the user has already revealed in their history, does this new detail significantly narrow the candidate population?”

Interaction examples: History “professional writer” + Draft “former paramedic” = rare career transition. History “London, college student” + Draft “works in a bookstore” = specific person. History “software developer” + Draft “native Arabic speaker” = narrow demographic.

Output: JSON with all attributes, new attributes, already known, cross-post interactions, draft risk alone, draft risk with history, and summary.

Step 3: Risk Analysis (Agent 2).

Task: Compute candidate set before and after the draft post.

Input: (1) The user’s existing profile (attributes from posting history, already public). (2) New attributes from the draft post.

Instructions:

- **Step A — Before draft:** Build a narrowing chain using only existing profile attributes. Start from the broadest reasonable population and apply attributes as filters.
- **Step B — After draft:** Extend the narrowing chain by adding new draft attributes. Show how each new attribute further narrows the candidate set.
- **Step C — Cross-post interactions:** For each new attribute, determine which existing profile attributes it interacts with to create multiplicative narrowing.

For each narrowing step, return: step, attribute, source (history/draft), population before, population after, narrowing factor, data source, reasoning.

Output: JSON with narrowing chain before, narrowing chain after, candidate set before, candidate set after, is tipping point, tipping point attribute, dangerous interactions, risk before, risk after, and recommendation.

Step 4: Phrase Ranking (Agent 3).

Task: Phrase-level risk ranking for the draft post.

Input: Draft text, existing profile from history, and risk analysis from Step 3 showing which draft attributes interact dangerously with history.

Instructions: Rank every identifying phrase in the draft by how much risk it adds, focusing on phrases that create cross-post interactions with the user’s history.

For each phrase, return: rank (1 = most dangerous), exact phrase, risk contribution (%), damage type (cross-post / standalone / sensitive), interacts with history, history post, removable (true/false), removal impact.

Key rule: Risk contributions reflect interaction value, not standalone value. “I was a paramedic” is low-risk alone but critical if history reveals “professional writer.” Percentages sum to ~100%.

Then model: (1) Immediate recognizers with specific descriptions, recognizing details, likelihood, and consequences. (2) Technical attack surface with exact search queries, platforms, expected result counts, and time estimates.

Output: JSON with phrase rankings, immediate recognizers, technical attack surface, cross-platform risk, and cross-platform reasoning.

Step 5: Edit Suggestion (Agent 4).

Task: Minimal privacy-preserving edits to break cross-post identification.

Input: Draft text, ranked phrases from Step 4, and risk context.

Key principle: Only the draft can be edited. History posts are already public. Edits must break the link between the draft and the history.

Edit types (in order of preference): (1) Generalize: make specific details less specific. (2) Fuzz: alter exact numbers slightly. (3) Rephrase: express the same idea with different words. (4) Remove: delete entirely (last resort).

For each edit, return: priority (1 = most impactful), original phrase, suggested edit, edit type, breaks interaction with, candidate set before, candidate set after, safety gain, what user loses, what user keeps, why this works.

Rules: Target phrases that create cross-post interactions first. Never edit emotional content, mental health details, or the core purpose of the post. The edited post must read naturally. Sometimes editing just 2–3 key phrases is enough.

Output: JSON with edits, edited post (full draft with all edits applied), original score, edited score, candidate set original, candidate set edited, edits count, words changed, story preserved, emotional tone preserved, core question preserved, and cross-post links broken.

Step 6: Verification (Agents 1 + 2).

Task: Re-run Steps 2 and 3 on the edited draft to confirm the cross-post links are broken.

Input: Edited draft from Step 5, plus the existing profile from Step 1.

Instructions: Agent 1 extracts attributes from the edited draft and checks for remaining cross-post interactions. Agent 2 recomputes the candidate set before and after. If the candidate set has expanded significantly (e.g., from 1 to 1,000+), the edits are confirmed effective. If risk remains high, the pipeline flags that additional edits are needed.

Output: Same format as Steps 2 and 3, plus a boolean risk reduced indicating whether the edits successfully broke the identification chain.